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# Visible-light mediated cascade cyclization of ene-vinylidenecyclopropanes: access to fluorinated heterocyclic compounds†

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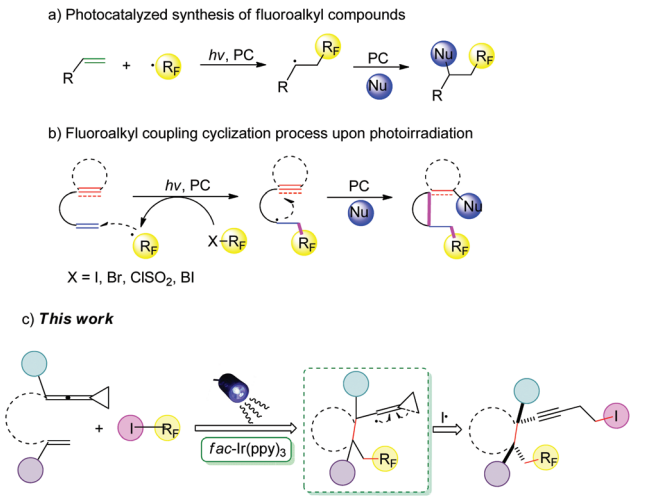
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A fluoroalkylation and cycloisomerization cascade reaction of ene-VDCP with ethyl iododifluoroacetate or perfluoroalkyl iodide upon visible-light irradiation is disclosed in this paper, affording the desired fluorinated cyclization products in moderate to good yields under mild reaction conditions along with good functional group tolerance. Mechanistic investigations were also performed with a light on/off experiment, fluorescence quenching experiment and quantum yield measurement, suggesting that a radical chain process exists in this process.

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Fluorinated heterocyclic compounds are important building blocks, which widely exist in bioactive molecules and natural products.[1] In addition, they are also core structural motifs in many synthesized organic compounds or pharmaceuticals. In fact, 30% of all agrochemicals approximately and 20% of pharmaceuticals currently on the market contain fluorine atoms.[1g] In general, introducing fluoroalkyl groups on heterocyclic stents can significantly enhance the pharmacokinetic properties of drugs by improving absorption, distribution, hydrophobicity and metabolism.[2] Therefore, it is very important to develop novel and efficient synthesis methods for heterocyclic compounds incorporating fluorine or fluorinated moieties. Thus, research on the synthesis methods of such compounds has always been a hot topic, attracting attention from organic chemists.[3]

In recent years, intramolecular tandem cyclization initiated by photocatalytic free radicals has been extensively reported. Through photoredox catalysis methods, simple alkenes can be utilized as feedstock materials to undergo the addition of fluoroalkyl radicals, generating fluorinated compounds *via* trifluoromethylation, iodinated perfluoroalkylation, thiodifluoroalkylation, and cyanofluoroalkylation (Scheme 1, a).[4] On the other hand, the addition of fluoroalkyl radicals generated from a fluoroalkane or other fluorinated precursors upon photoredox catalysis into alkenes can afford alkyl radicals, which can subsequently react with other intramolecular unsaturated functional groups, such as alkyne,[5] cyan,[6] aldehyde,[7] imine[8] and aromatic groups in an intramolecular manner,[9,10] to form fluorinated heterocyclic compounds (Scheme 1, b). However, only a few methods have been developed for the synthesis of fluorinated heterocyclic compounds through a tandem cyclization reaction under visible light irradiation thus far. For example, Wang's group[5a] reported a visible light-mediated bicyclization process employing ethyl bromofluoroacetate and 1,7-enyne for the synthesis of polycyclic compounds containing fluoroalkyl groups and heteroatoms. Moreover,




Scheme 1 Previous work and this work.

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†Electronic supplementary information (ESI) available: Experimental procedures and characterization data of new compounds. CCDC 2047670 for **2k**. For ESI and crystallographic data in CIF or other electronic format, see DOI: 10.1039/d1qo00540e



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Zhang's group[6] developed a visible-light-induced free radical cascade cyclization reaction of 2-isocyanoaryl sulfide with ethyl bromodifluoroacetate for the construction of 2-$CF_2$/$CF_3$-substituted benzothiazoles. At the present stage, the development of novel and atom/step economical tandem cyclization reactions for the rapid construction of fluorinated heterocylic products is highly desirable.

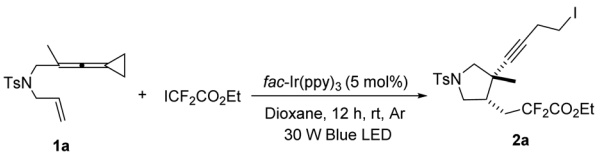
Although previous reports already demonstrated a wide range of unsaturated groups in olefinic substrates, the exploitation of new olefinic substrates is still desired. We attempted to introduce olefinic groups into vinylidenecyclopropanes (VDCPs)[11,12] that have been continuously studied by our research group, and hypothesized that ene-vinylidenecyclopropanes could be excellent candidates for exploiting new methods for fluoroalkylation and cyclization due to their multiple reaction sites. Herein, we wish to report a novel synthesis method for preparing fluorinated heterocyclic compounds through the generation of a fluoroalkyl radical upon photoredox catalysis by visible light irradiation and coupling with ene-VDCPs for a tandem ring opening/iodofluoroalkylation/cycloisomerization process (Scheme 1, c). The successful development of this new synthesis strategy can also enrich the chemistry of VDCPs.

Considering that homolytic cleavage of iodofluoroalkanes can be directly achieved by visible light irradiation, we initiated our investigation from the reaction between ene-VDCP **1a** and ethyl iododifluoroacetate under an argon atmosphere with blue LED irradiation at room temperature. After screening the solvent, lamp wattage and reaction time without adding a photocatalyst, we found that the desired product **2a** can be formed in 80% NMR yield (see Table S1 in the ESI†); however, the reaction always gave a complex mixture, from which it is difficult to get the pure product **2a**. Next, we tried adding a photocatalyst to the reaction system and conducting the reaction with 0.5 mol% *fac*-Ir(ppy)$_3$ as the photocatalyst and using dioxane as the solvent at room temperature upon irradiation with a 30 W blue LED for 12 h. We identified that the desired product **2a** could be isolated in 83% yield in pure form. Other photocatalysts such as Ir[dF($CF_3$)(ppy)]$_2$[dC$F_3$(bpy)]$PF_6$, Ir[dF($CF_3$)(ppy)]$_2$(dtbbpy)$PF_6$, Ir(ppy)$_2$(dtbbpy)$PF_6$ and Ir(dFC$F_3$ppy)$_2$(bpy)$PF_6$ turned out to be less efficient (Table 1, entries 2–5). The screening of solvents revealed that dioxane was more suitable than others (Table 1, entries 6–8).

Ethyl bromodifluoroacetate could also be used as the Rf radical precursor to react with **2a**, affording the desired product in 65% yield. However, when bromoperfluorohexane was utilized as the Rf radical precursor to react with **2a**, no reaction occurred (Table 1, entries 9 and 10). Control experiments demonstrated that no product **2a** was detected in the absence of visible light or under an ambient atmosphere (Table 1, entries 11 and 12).

With the optimized reaction conditions in hand, the generality of the production of fluorinated heterocyclic compounds **2** was explored using a series of substituted ene-vinylidenecyclopropanes as substrates and the results are shown in Scheme 2. With ene-VDCPs **1a–g** as the substrates ($R^1$ = H, $R^2$ = alkyl group), the desired products **2a–g** were obtained in 57%–83% yields as a single diastereoisomer and an increase of the steric hindrance of the alkyl group gradually decreased the yield of **2**. For ene-VDCPs **1h–j**, in which an aromatic ring ($R^1$ = Ar) was introduced, the desired products **2h–j** were obtained in 73–85% yields regardless of whether they are electron-neutral, -donating or -withdrawing aromatic rings. As for ene-VDCP **1k**, the corresponding product **2k** was formed in 50% yield presumably due to the steric effect of the cyclohexyl group. It is worth mentioning that the use of 1-butenyl group substituted ene-VDCP **1l** as the substrate afforded the desired product **2l** with a yield of 54%, indicating that the reaction has excellent functional group tolerance. In the case of substrate **1m**, in which the carbon chain of the olefinic moiety has been extended for another one carbon atom, the six-membered ring product **2m** was obtained in 43% yield. Obviously, compared with *N*-Ts-protected ene-VCDPs, *N*-Bs-protected substrates **1n–p** also performed very well, affording the desired products **2n–p** in 40–76% yields. Using ene-VDCP **1q** or **1r** connected by an O atom or a C atom as a substrate did not give the desired product **2q** or **2r** under the standard conditions. We speculate that when ene-VDCP was connected with an NTs moiety, the olefin and VDCP units were close to each other due to the Thorpe–Ingold effect in space, making the reaction proceed smoothly. However, when ene-VDCP was connected with a carbon or oxygen atom, the olefin and VDCP units could not be so close in space, ensuring that no reaction could take place or the reaction is difficult to occur, giving the desired product in low yield. The structure of **2k** was determined by X-ray crystallographic analysis and its ORTEP drawing is shown in Scheme 2.

**Table 1** Optimization of the reaction conditions for the synthesis of **2a**

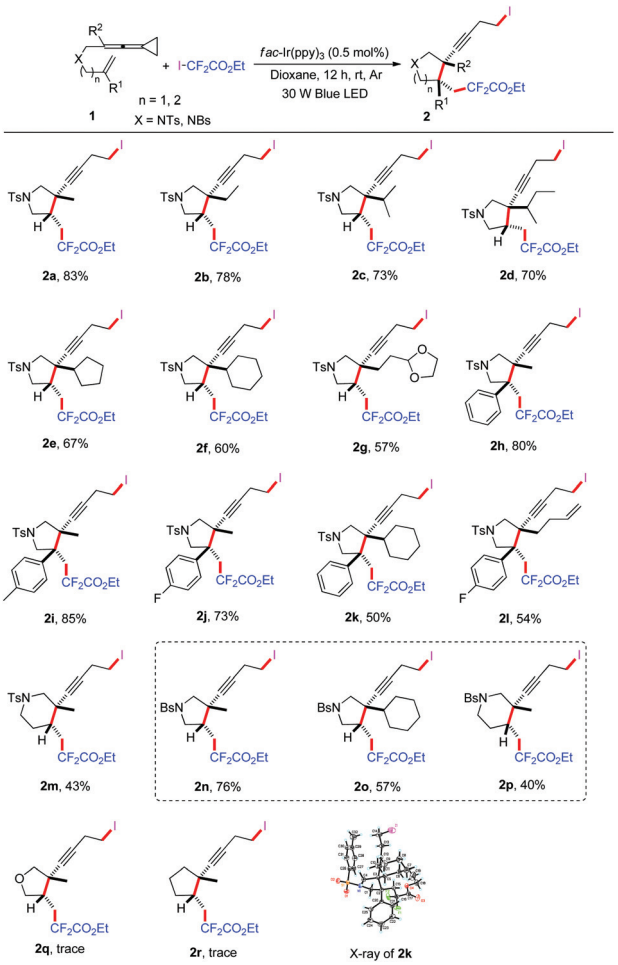



**1a** **2a**

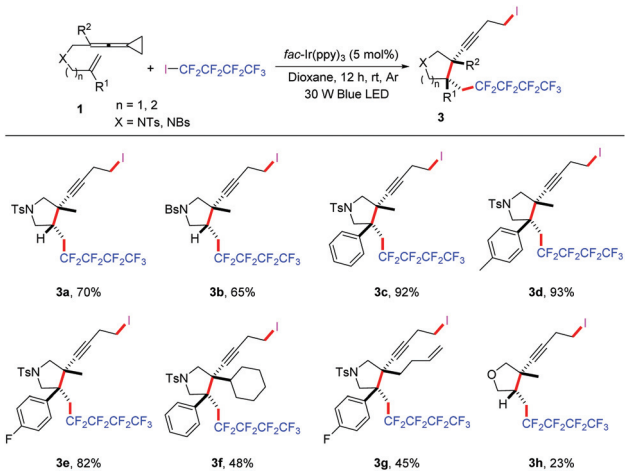
| Entry[a] | Variation from standard conditions | **2a**, yield[b] [%] |
|---|---|---|
| **1** | **None** | **92 (83)**[c] |
| 2 | Ir[dF($CF_3$)(ppy)]$_2$[dC$F_3$(bpy)]$PF_6$ as PC | 60 |
| 3 | Ir[dF($CF_3$)(ppy)]$_2$(dtbbpy)$PF_6$ as PC | 85 |
| 4 | Ir(ppy)$_2$(dtbbpy)$PF_6$ as PC | 35 |
| 5 | Ir(dFC$F_3$ppy)$_2$(bpy)$PF_6$ as PC | 51 |
| 6 | MeCN instead of dioxane | 48 |
| 7 | DCM instead of dioxane | 54 |
| 8 | DMF instead of dioxane | Trace |
| 9 | $BrCF_2CO_2Et$ instead of $ICF_2CO_2Et$ | 65 |
| 10 | $BrCF_2(CF_2)_4CF_3$ instead of $ICF_2CO_2Et$ | Nd |
| 11 | w/o light | Nd |
| 12 | Air instead of Ar | Nd |

[a] Reactions were carried out with **1a** (0.2 mmol, 1.0 equiv.) and the $ICF_2CO_2Et$ (0.24 mmol, 1.2 equiv.) photocatalyst (5 mol%) in solvents (10 mL) at ambient temperature under 30 W blue LED irradiation for 12 hours. [b] $^{19}F$ NMR yield using fluorobenzene as an internal standard. [c] Isolated yield.

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**Scheme 2** Substrate scope of products **2**. Reactions were carried out with **1** (0.2 mmol, 1.0 equiv.), $ICF_2CO_2Et$ (0.24 mmol, 1.2 equiv.), and *fac*-Ir(ppy)$_3$ (5 mol%), in dioxane (10 mL) at ambient temperature under 30 W blue LED irradiation for 12 hours. Yields were determined for isolated products.




**Scheme 3** Substrate scope of products **3**. Reactions were carried out with **1** (0.2 mmol, 1.0 equiv.), $ICF_2CF_2CF_2CF_3$ (0.24 mmol, 1.2 equiv.), and *fac*-Ir(ppy)$_3$ (5 mol%), in dioxane (10 mL) at ambient temperature under 30 W blue LED irradiation for 12 hours. Yields were determined for isolated products.

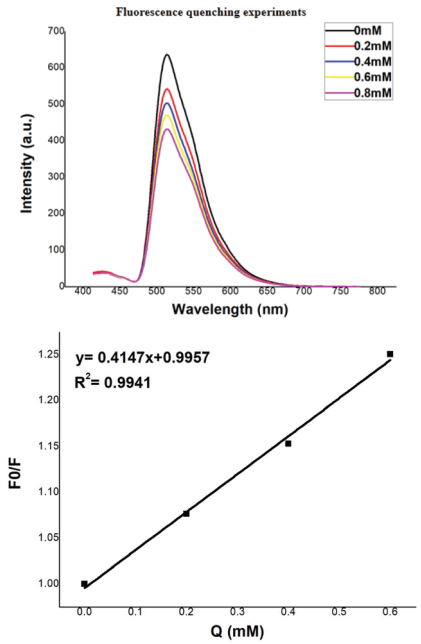
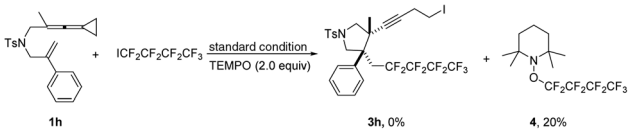



**Fig. 1** Fluorescence quenching experiment.




**Scheme 4** Radical trapping study.

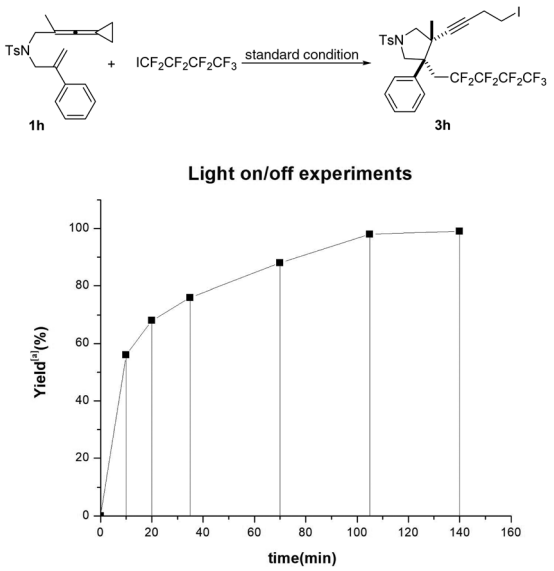
In order to further demonstrate the generality of this synthesis strategy, we then turned our attention to the scope of perfluoroalkyl iodides (Scheme 3). Similarly, using ene-VDCPs **1** as the substrates, the desired cyclic perfluoroalkylated products **3a–g** were obtained in moderate to good yields. Using methyl group-substituted VDCPs **1** ($R^1$ = H, $R^2$ = Me) as substrates gave the corresponding products **3a–b** in 65–70% yields.

Introducing aromatic substituents in VDCPs **1** ($R^1$ = Ar) gave the desired products **3c–e** in 82–93% yields. Further investigation indicated that the reaction also tolerated cyclohexyl and 1-butenyl groups ($R^2$ group), giving the corresponding products **3f** and **3g** in 48% and 45% yields, respectively. When ene-VDCP connected by an O atom was used as the substrate to react with iodoperfluorobutane, the desired product **3h** was obtained in 23% yield.

To gain more insights into the reaction mechanism, we conducted several control experiments (Fig. 1 and Schemes 4 and 5).[13] First, when the radical scavenger 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) (2.0 equiv.) was added to the reaction system, the reaction was completely inhibited

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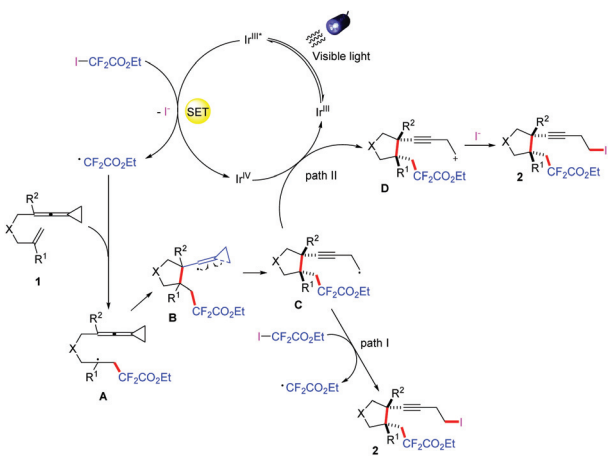
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Scheme 5 Light on/off experiments.

under the standard conditions and the corresponding TEMPO-incorporated perfluoroalkyl radical was isolated in 20% yield (Scheme 4). Fluorescence quenching analysis revealed that $ICF_2CO_2Et$ was an effective quencher for the excited-state photocatalyst (Fig. 1). Next, to clarify whether the reaction also proceeded through a radical chain process,[14] we performed light on/off experiments with **1h** as a substrate, disclosing that the reaction also proceeded when the light was turned off (Scheme 5). Furthermore, the quantum yield was measured as $\Phi = 23$ in this reaction, making the intervention of the radical chain mechanism likely (see the ESI† for details).

On the basis of the mechanistic investigations discussed above as well as the related previous reports,[15] a plausible reaction mechanism was proposed, as shown in Scheme 6. Irradiation of *fac*-$Ir^{III}(ppy)_3$ with blue light generates the excited-state of *fac*-$Ir^{III*}(ppy)_3$, which undergoes oxidative quenching with ethyl iododifluoroacetate through a SET process to give an electrophilic fluoroalkyl radical, tending to react with the electron-rich alkene moiety VDCP **1** to furnish a radical intermediate **A**. Then, intramolecular cycloaddition takes place to afford radical intermediate **B**, which undergoes a cyclopropane ring-opening reaction to give alkynyl radical intermediate **C**. There are two pathways to obtain the desired product **2**. Path I is a radical chain process, where the radical **C** reacts with ethyl iododifluoroacetate to afford product **2** with the regeneration of the fluoroalkyl radical. Another pathway, path II, involves the oxidation of radical **C** by $Ir^{IV}$ to afford cationic intermediate **D**, which subsequently undergoes nucleophilic addition of an iodide anion also to generate the final product **2**.




Scheme 6 Proposed mechanism for the production of **3a**.

In summary, we have developed a novel protocol for visible-light-mediated carbocyclization reactions of ene-VDCPs with ethyl iododifluoroacetate and perfluoroalkane for the synthesis of fluorinated heterocyclic compounds under mild conditions. The reaction had good functional group tolerance and substrate scope, affording the desired products in moderate to good yields. These five or six-membered fluorine-containing heterocyclic compounds may be useful as potential intermediates in organic synthesis and medicinal chemistry. The mechanistic study indicated that the reaction involved a free radical chain process and a sequential fluoroalkylation and cycloisomerization process. Further investigations on expanding the reaction of this synthesis method are ongoing in our laboratory.

## Conflicts of interest

There are no conflicts of interest to declare.

## Acknowledgements

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